

# Jonnathan Singh Alvarado, Ph.D.

## Contact

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jonnasingh.github.io

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## Education

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Postdoctoral Fellow, Neurobiology, Harvard University 2021-Current

Ph.D., Neurobiology, Duke University 2014-2021

B.S., Microbiology, University of Florida 2009-2013

## Research Experience

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Postdoctoral Fellow 2021-Current

Dr. Mark Andermann, Beth Israel Deaconess Medical Center

Graduate Student, Duke University 2014-2021

Dr. Richard Mooney, Department of Neurobiology

Research Assistant, Duke University 2013-2014

Dr. Cagla Eroglu, Department of Cell Biology

Undergraduate Researcher, University of Florida 2012-2013

Dr. Stephanie Karst, Department of Virology

Undergraduate Researcher, University of Florida 2011-2012

Dr. Brent Reynolds, Department of Cancer Biology

## Publications

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EASI-PASS: an accessible pipeline for linking functional imaging and mRNA profiling. **Singh Alvarado J**, Massengill C, Stern J, Amsalem O, Ventura B, Jang A, Cook S, Veliche A, Sunkavalli P, Patel D, Colaccino J, Evans K, Wang Y, Andermann ML. *bioRxiv*. 2026.

The role of modern neuroscience in the renaissance of psychedelic research. Evans KE, Massengill C, **Singh Alvarado J**. *Revista Costarricense de Psicología*. 2025.

A sensitive soma-localized red fluorescent calcium indicator for in vivo imaging of neuronal populations at single-cell resolution. Zhou S, Zhu Q, Eom M, Fang S, Subach O, Ran C, **Singh Alvarado J**, Sunkavalli P, Yuanping Dong, Wang Y, Wang Z, Sun X, Fedor V Subach, Kiryl D Piatkevich. *PLoS Biology*. 2025.

Transient cAMP production drives rapid and sustained spiking in brainstem parabrachial neurons to suppress feeding. **Singh Alvarado J**\*, Lutas A\*, Madara JC, Isaac J, Lommer C, Massengill C, Andermann ML. *Neuron*. 2024.

Divergent projections from Locus Coeruleus to the cortico-basal ganglia system and ventral tegmental area of the zebra finch. **Singh Alvarado J**\*, Hatfield J\*, Mooney R. *Journal of Comparative Neurology*. 2022.

Neural dynamics underlying birdsong practice and performance. **Singh Alvarado J**, Goffinet J, Michael V, Liberti W, Hatfield J, Gardner T, Pearson J, & Mooney R. *Nature*. 2021.

Building Opportunities and Overtures in Science and Technology: Establishing an Early Intervention, Multi-level, Continuous STEM Pathway Program. Garcia J, Gibson R, Breslav A, **Singh Alvarado J**, Seas A, Maduka A, et al. *Journal of STEM Outreach*. 2021.

MIN1PIPE: A Miniscope 1-Photon-Based Calcium Imaging Signal Extraction Pipeline. Lu J, Li C, **Singh Alvarado J**, Zhe Charles Zhou, Flavio Fröhlich, Mooney R, Wang F. *Cell Rep*. 2018.

Expressing mutant huntingtin in the songbird basal ganglia disrupts cortico-basal ganglia networks and vocal sequences. Tanaka M, **Singh Alvarado J**, Murugan M, and Mooney R. *Proc. Natl. Acad. Sci. USA*. 2016.

Astrocytes refine cortical connectivity at dendritic spines. Risher WC, Patel S, Kim IH, Uezu A, Pilaz LJ, **Singh Alvarado J**, Calhan OY, Silver DL, Stevens B, Soderling SH, and Eroglu C. *eLife*. 2014.

Rapid Golgi Analysis Method for Efficient and Unbiased Classification of Dendritic Spines. Risher WC, Ustunkaya T, **Singh Alvarado J**, Eroglu C. *PLoS ONE*. 2014.

\* denotes equal contribution.

## Talks

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|---------------------------------------------------------------------------------------|------|
| Outstanding Postdoc Talk – HMS Neurobiology Departmental Retreat                      | 2026 |
| Optogenetic Approaches to Understanding Neural Circuits and Behavior – Lucca, Italy   | 2026 |
| Systems Neuroscience Club – Harvard Medical School                                    | 2026 |
| Departmental Pizza Talks – Harvard Medical School Department of Neurobiology          | 2025 |
| Department Lab Results Seminar – Boston Children’s Hospital Kirby Neurobiology Center | 2024 |
| Endocrinology Division Meeting – BIDMC                                                | 2024 |
| Special Seminar – University of Pittsburgh & Carnegie Mellon University               | 2024 |
| SFN Minisymposium – Peripheral mechanisms of energy & metabolism                      | 2023 |
| Special Seminar – Champalimaud Centre for the Unknown                                 | 2023 |
| Endocrinology Division Meeting – BIDMC                                                | 2022 |
| Joint Lab Meeting – Carnegie Mellon University Systems Club                           | 2022 |
| COSYNE (6% of abstracts selected)                                                     | 2021 |
| Special Seminar – Columbia University                                                 | 2021 |
| Virtual Dopamine Conference (ViDA)                                                    | 2021 |
| University of Washington and Allen Institute Neuromodulation Club                     | 2020 |
| Neuromodulation Gordon Research Conference                                            | 2019 |
| Duke University Brown Bag Seminars                                                    | 2018 |
| Duke Institute for Brain Sciences Symposium                                           | 2017 |
| Duke TEDx                                                                             | 2016 |

## Fellowships and Awards

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|---------------------------------------------------------------------|-----------|
| BBRF Young Investigator Grant, Brain & Behavior Research Foundation | 2026-2028 |
| David Hubel Outstanding Postdoc Award                               | 2026      |
| The Transmitter “Rising Stars of Neuroscience”                      | 2025      |
| NINDS K99/R00 (parent mechanism), deferred                          | 2025      |
| MOSAIC NINDS K99/R00 Award                                          | 2024      |
| Harvard Brain Initiative Postdoctoral Pioneer Award                 | 2024-2025 |
| Harvard Mind, Brain, and Behavior (MBB) postdoctoral fellowship     | 2022-2024 |
| Graduate student training enhancement grant (GSTEG) awardee         | 2019      |
| Simons Foundation Neuromodulation GRC – Outstanding Talk Award      | 2019      |
| Cold Spring Harbor Laboratories – Neural data science course        | 2019      |
| Duke Institute for Brain Sciences Symposium – Best Talk             | 2017      |
| Ruth K. Broad Research Fellowship                                   | 2016      |
| Biosciences Collaborative for Research Engagement (BioCoRe) Scholar | 2015      |
| Latin American & Caribbean Scholarship (4-year scholarship)         | 2009-2013 |

## Leadership

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|----------------------------------------------------------------------------|--------------|
| <b>Gordon Research Seminar Co-chair</b>                                    | 2025-Present |
| Organizer for the 2027 Neuromodulation Gordon Research Seminar, Barcelona. |              |

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| <b>HPREP mentor</b>                                                                   | 2024-Present |
| Lecturer and hands-on project coordinator for STEM outreach program HPREP at Harvard. |              |

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| <b>NeuroLaunchpad</b>                                                                                                                                           | 2021-2022 |
| Created and received funding for a trainee-centered nationwide seminar series to give exposure to burgeoning scientists, resulting in over 1000 unique viewers. |           |

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| <b>BOOST Coach</b>                                                                                                                 | 2014-2021 |
| Developed year-long STEM-based research programs and hands-on learning curriculums for low-income middle and high school students. |           |

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| <b>Committee for Reimagining Doctoral Education at Duke</b>                                                                                                                                         | 2018-2020 |
| One of three Duke graduate students selected to serve. Met monthly with 30 faculty, the provost, and the research training Dean to construct a strategic plan for restructuring doctoral education. |           |

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| <b>Other Service</b>                                                                                                                                                                                               | 2014-2021 |
| Neurobiology Diversity and Inclusion taskforce, Neurobiology Steering Committee, Deans Mentoring and Gordon G Hammes mentoring award committees, Faculty advancement committee, Alfred P Sloan Scholars committee. |           |

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| <b>Rex Scholar</b>                                                                                  | 2016-2017 |
| Assisted in developing software and virtual lectures to be widely distributed in NC public schools. |           |

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| <b>TECHO – Costa Rica NGO</b>                                                                          | 2007-2011 |
| Directed teams of volunteers in the construction of emergency transitional homes for families in need. |           |